No.

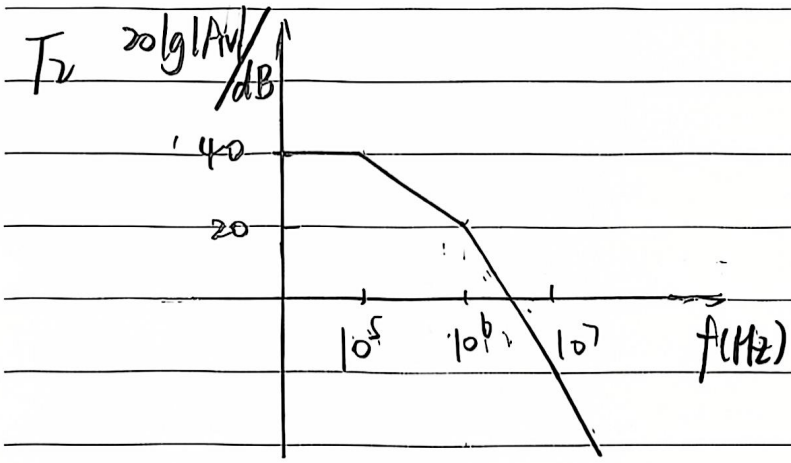
Date.

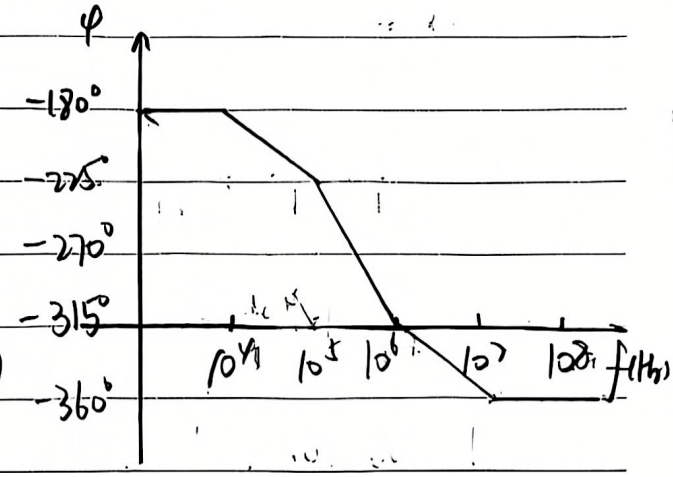
第8次作业

C10P1

T1 C

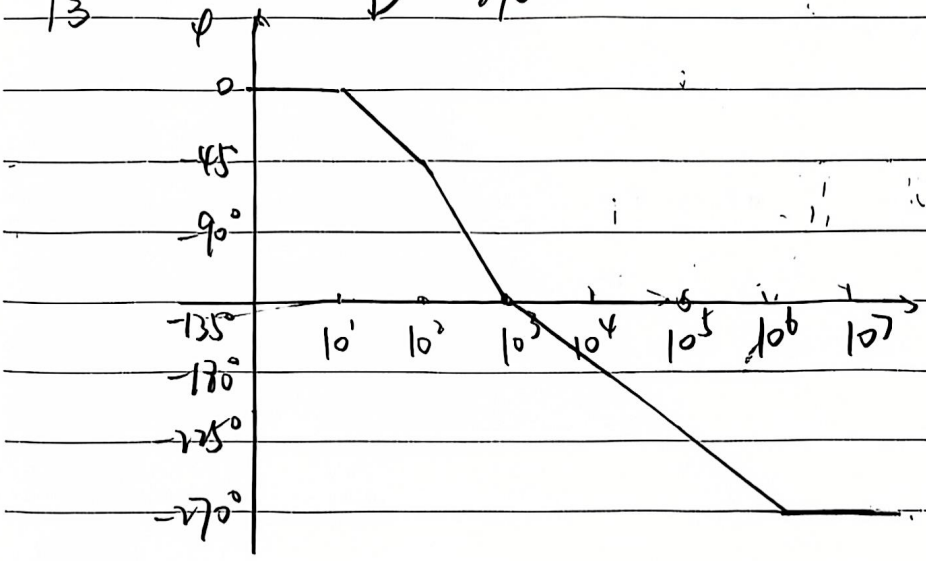
T2 $20\lg|A_{VM}|$/dB





$f_H \approx 100kHz$

T3 D -270°



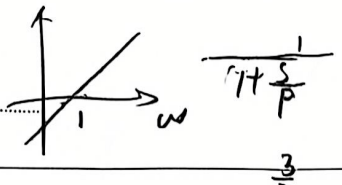
T4 $A_V = \dfrac{10^4}{(1+\frac{jf}{10^5})(1+\frac{jf}{10^7})(1+\frac{jf}{10^8})}$

T5 -45° 45°

No.

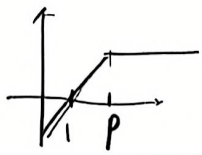
Date.

S:



$\frac{1}{1+\frac{s}{p}}$





$\frac{s}{1+\frac{s}{p}}$ ——

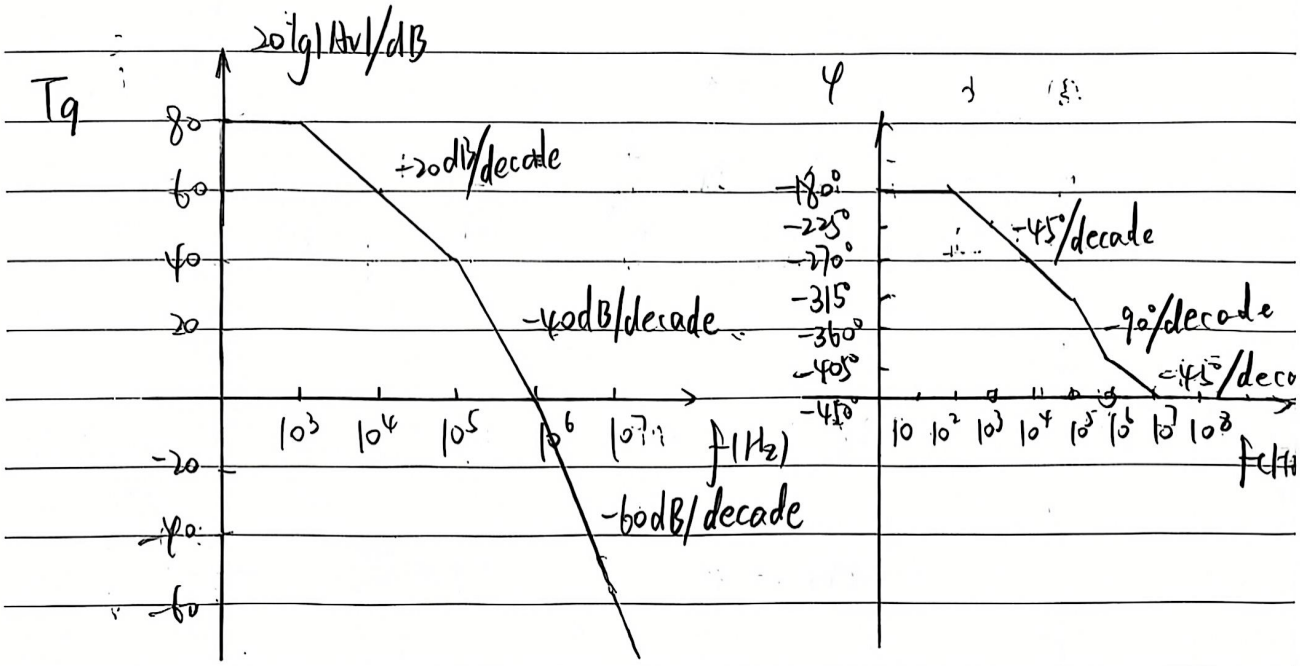
T6 $A_V = \frac{-10^{\frac{3}{2}}}{(1+\frac{jf}{10^5})(1+\frac{10}{jf})}$

$\frac{1}{1+\frac{p}{s}} = \frac{s}{s+p} = \frac{1}{p}\frac{s}{1+\frac{s}{p}}$

T7 $f_L = 10Hz$ $f_H = 10^5 Hz$

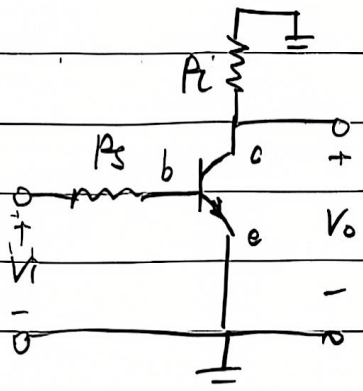
T8 $A_V = \frac{10^3}{(1+\frac{jf}{10^4})(1+\frac{s}{10^5})(1+\frac{10}{s})}$

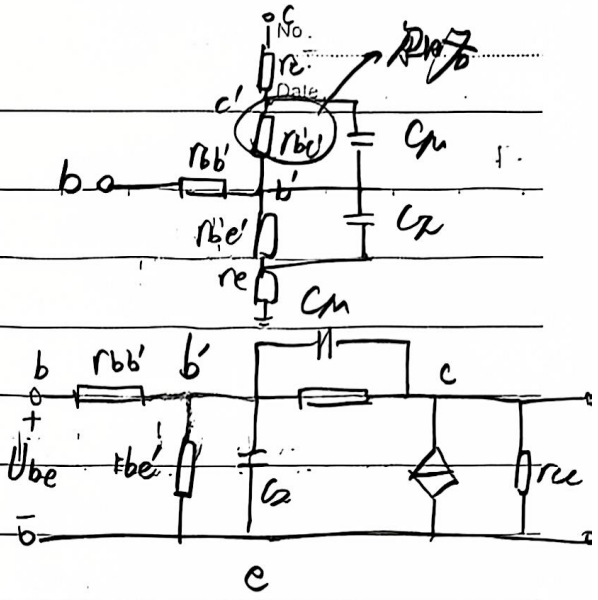
T9

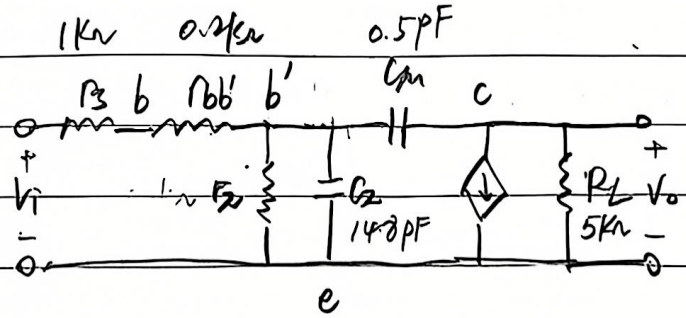


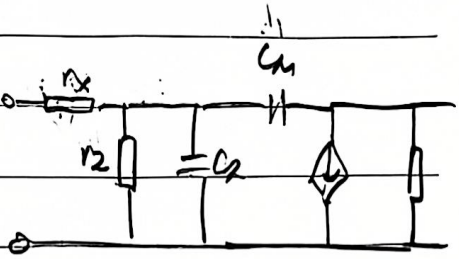
C10P2

T1







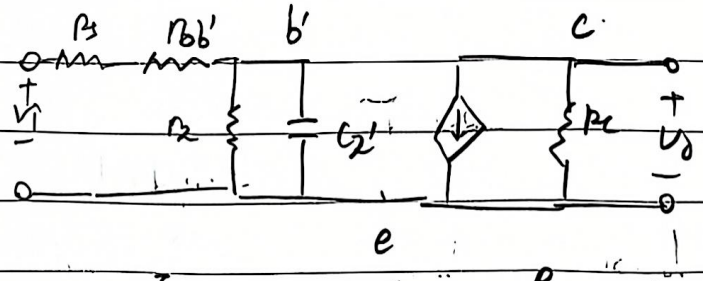


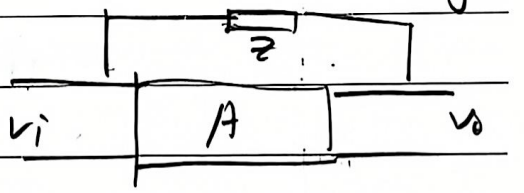
3700

$I_{EQ}=1mA$

$V_T=26mV$

Cin主导 $\omega_H=\frac{1}{R'_{sig}C_M}$





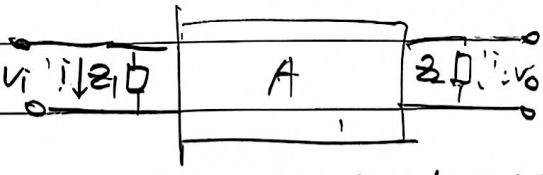
$g_m=\frac{I_{EQ}}{V_T}=38.5mS$ $r_\pi=\frac{\beta}{g_m}=2.60K\Omega$

$C'_2=C_2+(1+g_mR'_L)C_\mu=111.55pF$

$R'_{sig}=(R_s+r_{bb'})\|R_2=0.82K\Omega$

$\omega_H=\frac{1}{C'_2R'_{sig}}=1.09\times10^7\ rad/s$

$f_H=\frac{\omega_H}{2\pi}=1.73\times10^6Hz$



$I=\frac{V_i-V_o}{Z}=\frac{V_i(1+A)}{Z}=\frac{V_i}{\frac{Z}{1+A}}=\frac{V_i}{Z_i}$

$Z_i=\frac{Z}{1+A}\Rightarrow C_i=(1+A)C$

T2 B. A [耦合，分布？]

$I=\frac{V_i-V_o}{Z}=\frac{\frac{A+1}{A}V_o}{Z}=\frac{V_o}{\frac{A}{A+1}Z}$

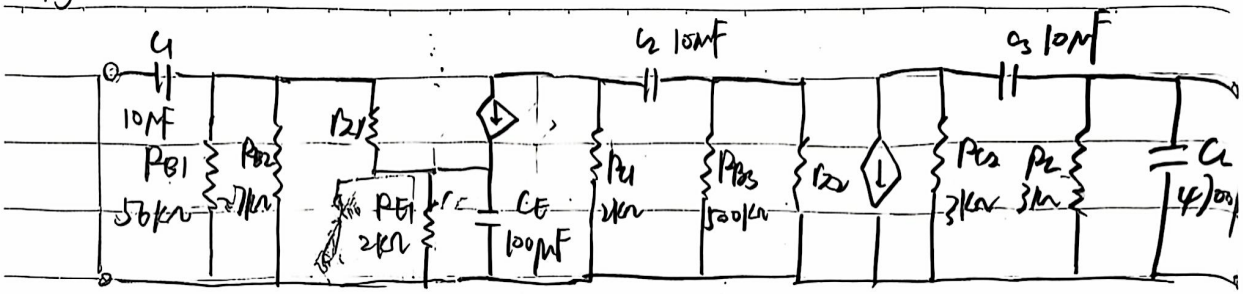
$Z_o=\frac{A}{A+1}Z$ $C_o=(1+\frac{1}{A})C$

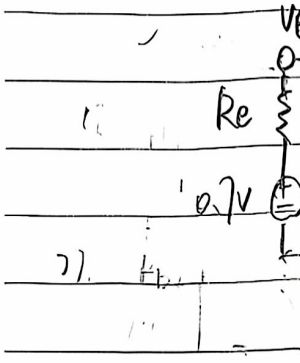
No

Date

$Z_C = \frac{1}{j\omega C}$ 电容↑隔直 C↑ Z_C↓通交↑

T3





$\therefore V_{BE} = 1.627V$, $R_e = 18.2k\Omega$

$I_b R_e + 0.7 + (1+\beta) I_b R_{E1} = V_{BE}$

$\Rightarrow I_b = 4.21\mu A$, $I_c = \beta I_b = 0.421mA$

$r_{be} = \frac{\beta V_T}{I_e} = 6.12k\Omega$

$I_{b2} = 8.6\mu A$, $I_{c2} = \beta I_{b2} = 0.86mA$, $r_{be2} = 3.02k\Omega$

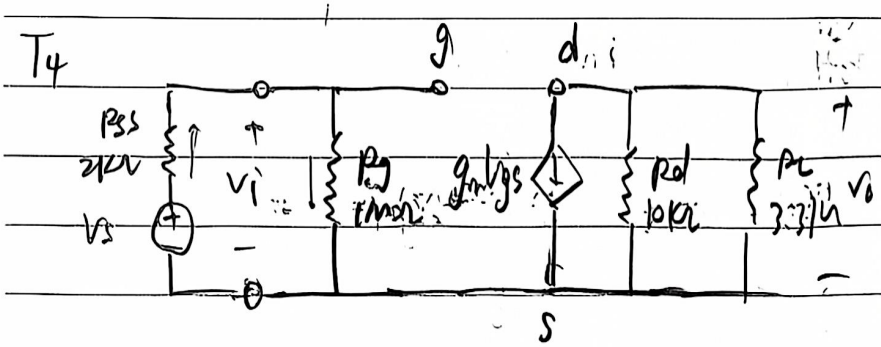
$\therefore R_1 = R_{B1} // R_{B2} // r_{be}$, $f_{c1} = \frac{1}{2\pi C_1 R_1} = 3.47Hz$

$R_E = R_{E1} // \frac{r_{be2}}{1+\beta}$, $f_{CE} = \frac{1}{2\pi C_E R_E} = 27.06Hz$

$R_2 = R_{e1} + (R_{B3} // r_{be2})$, $f_{c2} = \frac{1}{2\pi C_2 R_2} = 3.183Hz$

[C和1+并联] $R_3 = R_{e2} + R_L$, $f_{c3} = \frac{1}{2\pi C_3 R_3} = 2.65Hz$

T4



$$A_{vm} = \frac{V_o}{V_S} = \frac{-g_m V_{gs}(R_d // R_L)}{V_{gs} + \frac{V_{gs}}{R_G} R_{SS}} \rightarrow -12.38$$

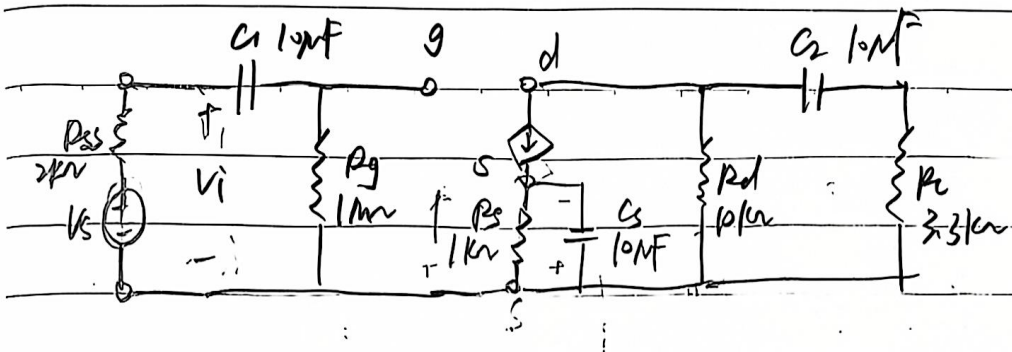
$C_{in} = (1+A_m) C_{gd} + C_{gs} = 71.9pF$ $R_{sig} = R_G // R_{SS} = 1.996k\Omega$

$f_H = \frac{1}{2\pi C_{in} R_{sig}} = 1.109MHz$

$g_m V_{gs}$

No.

Date.



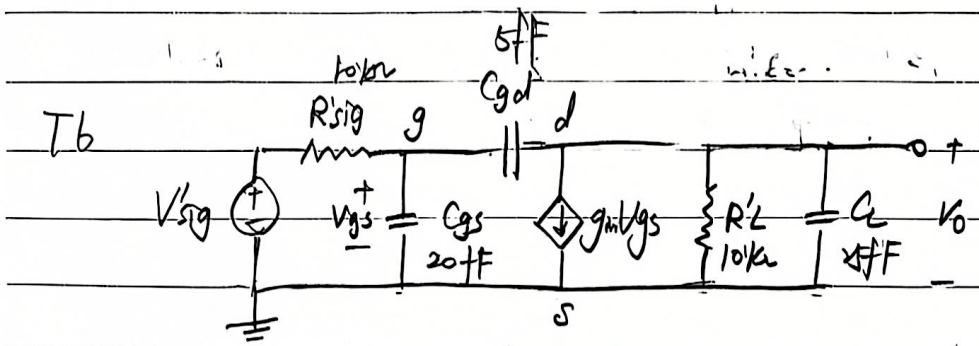
$R_1 = R_g + R_{ss}$, $f_{c1} = \frac{1}{2\pi C_1 R_1} = 0.0159\,Hz$

$R_2 = R_d + R_L$, $f_{c2} = \frac{1}{2\pi C_2 R_2} = 1.197\,Hz$ 故 $f_L = 95.49\,Hz$

$R_{es} = \frac{V_{gs}}{g_m V_{gs} + \frac{V_{gs}}{R_s}}$, $f_{cs} = \frac{1}{2\pi C_s R_{es}} = 95.49\,Hz$

T5 $\xrightarrow{I_i}$

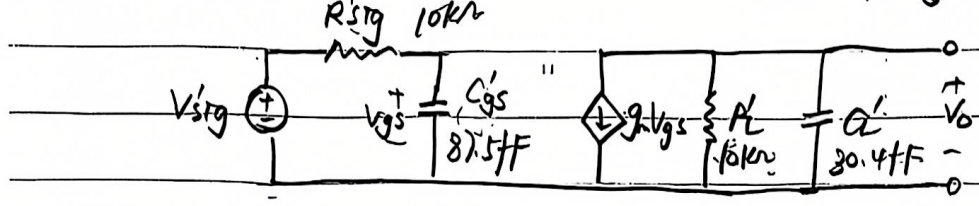
D C R i_o ? $A_v = \frac{i_o}{i_n} = \frac{i_o}{\frac{i_o R}{\frac{1}{j\omega C}} + i_o} = \frac{1}{j\omega RC + 1} \to 1$ $\omega \to 0$



$g_m = 1.25\,mA/V$

$A_v = \frac{-g_m V_{gs} R'_L}{V_{gs}} = -12.5$ $C_1 = (1+A)C_{gd} = 67.5fF$

$C_2 = (1+\frac{1}{A})C_{gd} = 5.4fF$



$f_{H1} = \frac{1}{2\pi R'_{sig} C_{gs}} = 1.82 \times 10^8\,Hz$ $f_{H2} = \frac{1}{2\pi R'_L C'_L} = 5.24 \times 10^8\,Hz$

$f = \frac{1}{\frac{1}{f_{H1}} + \frac{1}{f_{H2}}} = 1.35 \times 10^8\,Hz$

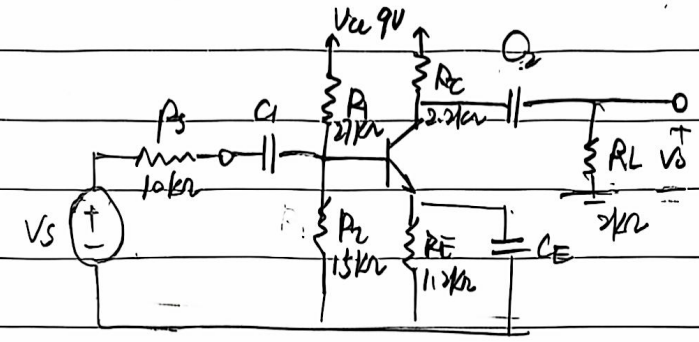
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T7 高频 附频 (高3)

T8



$\beta=100$, $V_{BE}=0.7V$

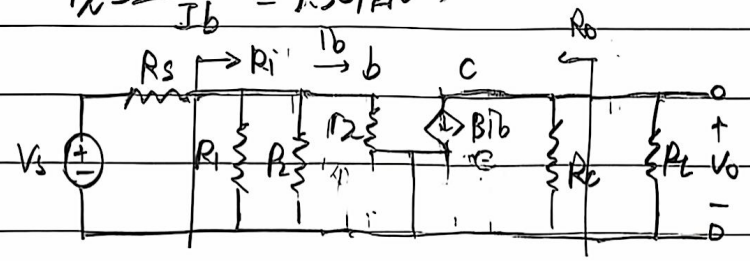
$V_T=25mV$

(1) $V_{be}=3.214V$, $R_{be}=R_1 \| R_2=9.643K\Omega$

$I_bR_{be}+V_{BE}+(\beta+1)I_bR_E=V_{be} \Rightarrow I_b=0.0192mA$

$I_E=(\beta+1)I_b=1.9406mA$

(2) $r_\pi=\frac{V_T}{I_b}=1.301K\Omega$



$R_i=R_1 \| R_2 \| r_\pi=1.146K\Omega$

$R_o=R_c=2.2K\Omega$

$\frac{V_o}{V_s}=\frac{\beta I_b(R_c \| R_L)}{I_b r_\pi}\cdot\frac{1}{\frac{R_i+R_s}{R_i}}=-8.279$

(3) $R_{CE}=R_E \| (r_\pi+R_1 \| R_2 \| R_s)=1.006K\Omega$

$6.210K\Omega$

$f=\frac{1}{2\pi R_{CE}C_E}=15.82Hz$

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$\times 10^{20}$ No.

Date.

20dB ← 10倍

6dB ← 2倍

C10P3

T1 $(1+A_0)CF$ $(1+\frac{1}{A_0})CF$

T2 → 3dB → 0.708倍